iMS Robot Management System User Manual

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Beijing Huiwen Technology (Group) Co., Ltd.

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Version Record

Revised Date	Version Number	Update Content	Editor
20250226	V0.5.0	The first version of the iMS User Manual	Jiurui Ye
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20250722	V1.1.0	Add function descriptions of remote images and new function modules.	Jiurui Ye
20250820	V1.1.5	Adapt to the function description of the online version of iMS Cloud 1.1.5	Jiurui Ye

1 Overview

1.1 Welcome

Welcome to read the User Manual of iBEN ROBOT Management System (iMS).

iBEN ROBOT Management System is a professional business process management software that can manage and schedule robots with SLAM navigation for automated handling. It replaces the heavy manual labor by efficiently handling the automatic flow of robots and shelves between multiple locations, and can be widely used in different industries such as manufacturing, warehousing and logistics to provide professional production logistics management solutions.

1.2 Statement

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- Error name

- Version
- Questions/Difficulties
- Suggestions and revision comments

This document is an approved robot management system user manual.

To view the latest iMS manual, please visit https://www.ibenrobot.com/ to contact the company's technical support.

Reference:

Some illustrations in this manual are intended to help understand the concepts and use of the system and may differ from the actual product.

Please keep this manual properly for future reference.

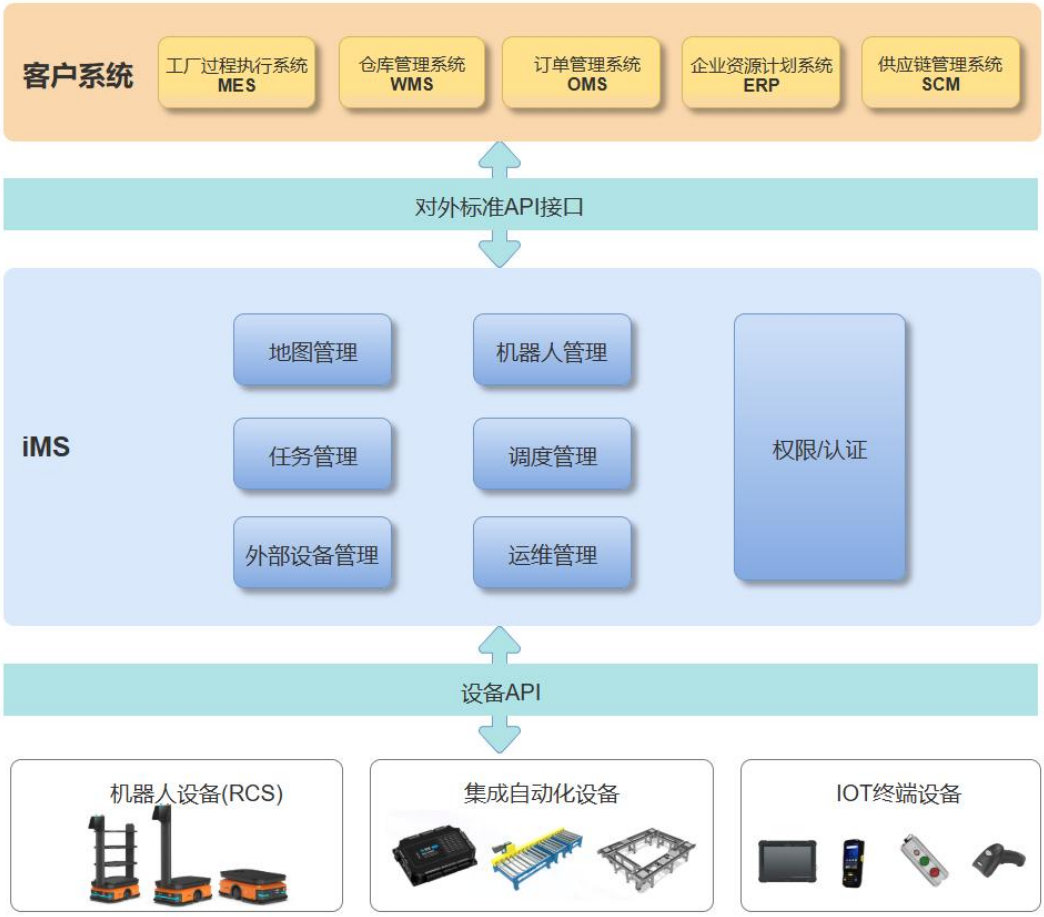
1.3 Intentions of This Manual

This manual introduce the various functions of iMS and describes the business processes and system operation methods. This document is wrote and revised by product and technical personnel, and finally approved. It serves as a technical guidance document for subsequent product training and implementation training for internal employees and external customers.

2 Function Description

2.1 System Architecture

2.1.1 System Architecture Diagram



2.1.2 Version Information

The entry is located at the upper right corner of the platform. Click the question mark icon to view the version information of the current iMS.

2.1.3 User Information

User accounts are assigned by the enterprise administrator, including the account name and initial password. Users can change the password. If the password is forgotten, they can contact the superior super administrator to reset the password.

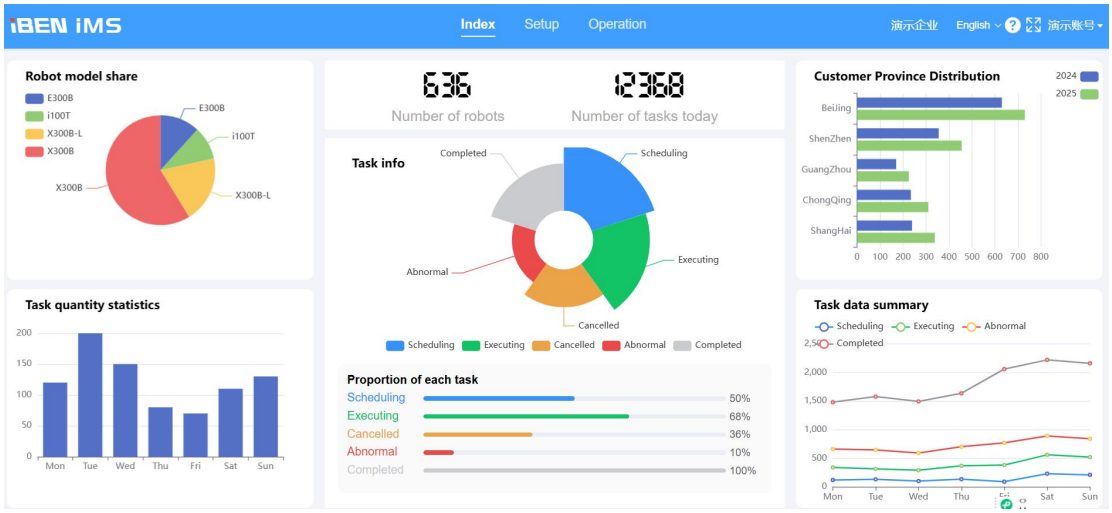
ⓘ There are 2 special roles preset in the system: super administrator and enterprise administrator. When opening an enterprise account, you can select the permission of the enterprise administrator role. The permission control of the enterprise administrator role has been configured when the system is launched, and some functions cannot be used. The enterprise administrator is the user with the highest permission in the enterprise, and can create sub-accounts and sub-roles under each enterprise respectively.

2.1.4 Internationalization Adaptation

Currently, Chinese and English language are supported. The language can be

switched on the login page and the upper right corner of the homepage.

2.2 Homepage



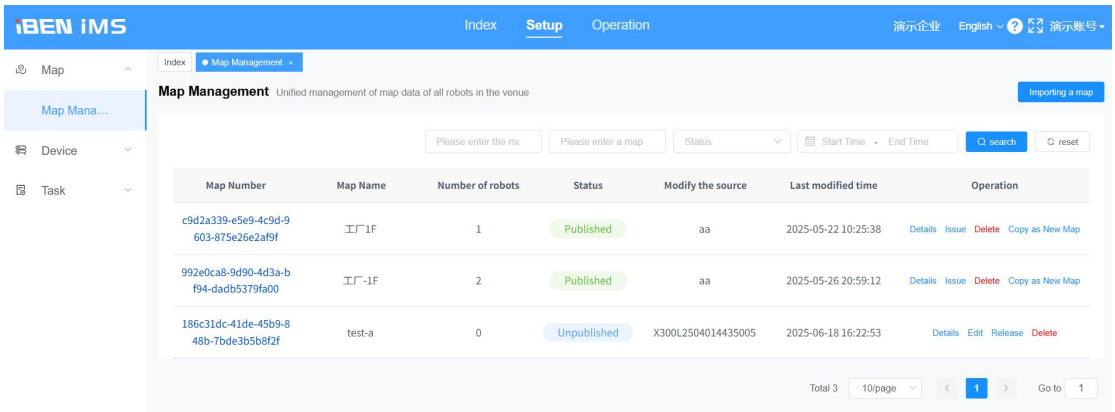
The homepage is the first-level menu of the entire system, which displays the visual statistical data of the entire system, including task execution status, robot model proportion, robot status and other data.

2.3 Configuration

The configuration menu is the first-level menu of the entire system, which manages all functions under the entire enterprise that require configuration operations.

2.3.1 Map

Manage map data of all robots in the site.



2.3.1.1 Map Status and Operation Permissions

Status	Description	Operation Permissions
Unpublished	Newly created / imported draft map or edited version.	Allows editing, publishing, deleting, and specifying robot operation.
Published	The official version that takes effect and can be issued to all robots	Allows issuing, copying as a new map, and deleting (only if there are no bound robots)

2.3.1.2 Map Source

The "Import Map" button in the cloud platform map list or the "Upload to cloud platform" button on the robot side.

2.3.1.3 Publish Map

Unpublished maps need to be manually clicked "Publish".

After publishing, the status changes to "Published", and the previously published version of the map will be automatically overwritten.

The new version is visible to all robots and supports “Issue”.

2.3.1.4 Copy Map

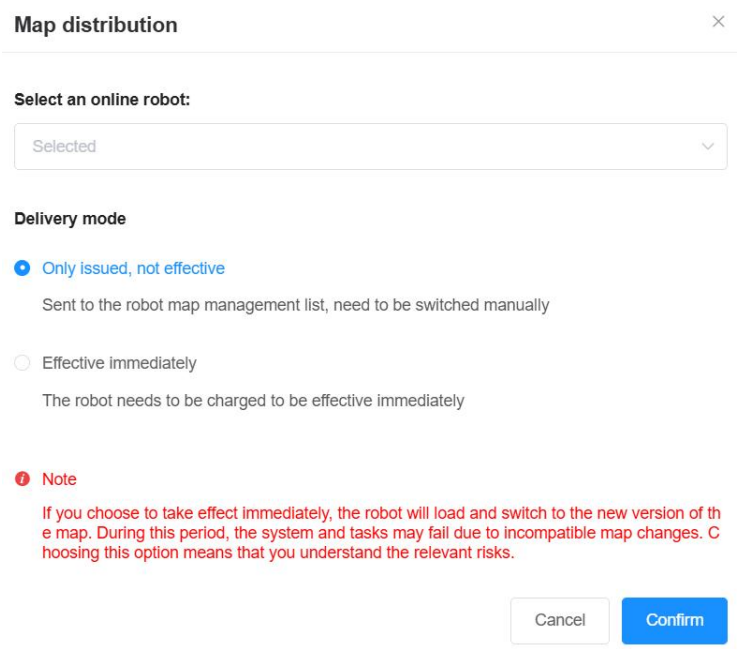
- Scenario: When a new version needs to be created based on a published map, click "Copy as a new map".
- Result: A copy in the unpublished state is generated, the map number is updated, and the map name is changed to a new name. The rule is: Name + copy + time. The copied map can be edited independently and then republished.

2.3.1.5 Delete Map

- Unpublished map: Directly click "Delete" (The number of bound robots must be 0).

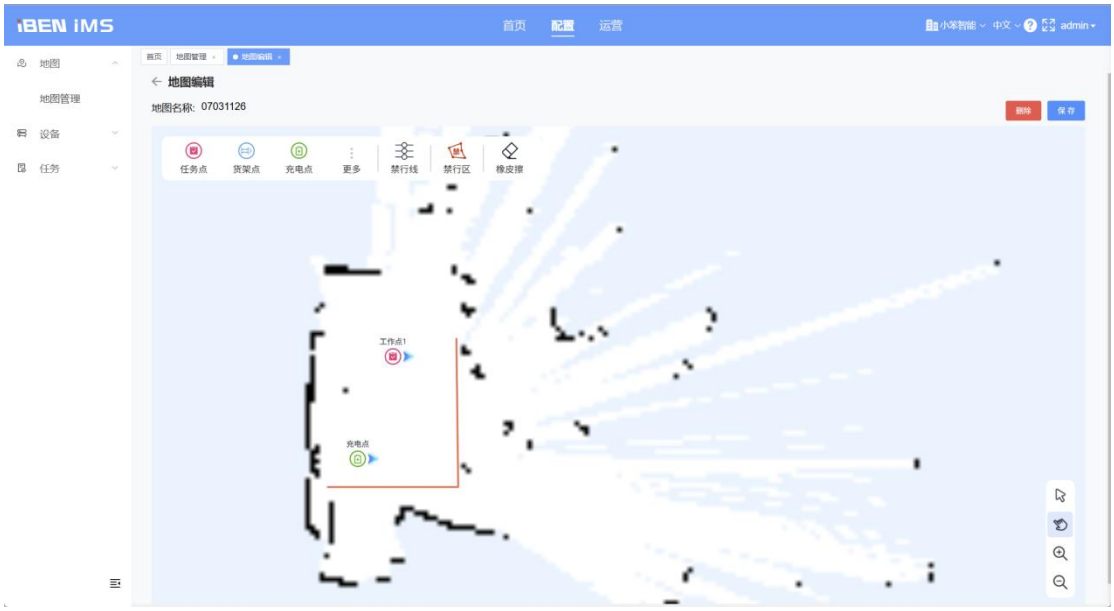
- Published map: All robots must be unbound first (ensure no task reply on) before deletion.

2.3.1.6 Issue Map



2.3.1.7 Edit Map

Currently, it can edit the functions such as waypoint, no-entry line, area, eraser, including the direction of waypoint and other modules.



2.3.2 Equipment

Manage all equipment connected with the iMS system.

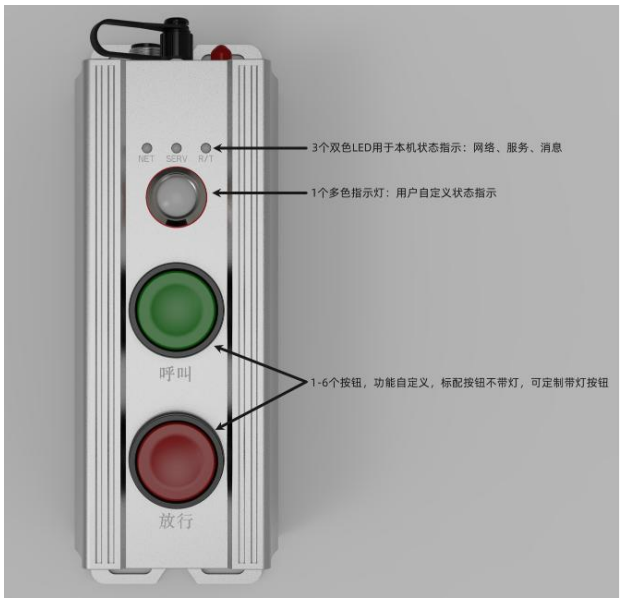
2.3.2.1 Pager Management

Management of pagers and configuration of tasks bound to pagers.

After the pager leaves the factory, it needs to be configured to connect with the cloud platform. The cloud platform can add a new pager and enter the pager number that has successfully connected to the cloud platform to obtain the pager's data and conduct two-way communication. For the complete configuration operation of the pager accessing the cloud platform, please refer to the pager configuration document.

2.3.2.2 Pager Introduction

Pager Example Picture



LED Indicator	Status	Description
NET (Network)	Red light blinking	WIFI network connecting
	Green steady on	WIFI network connected successfully
SERV (Service)	Red light blinking	Socket connecting or creating
	Green steady on	In TCP Server mode, there is at least one client connected successfully; In TCP Client mode, the device has been

		connected to the server; In UDP mode, the Socket has been created successfully.
R/T (Transmit/Receive)	Red light blinking	Blinks once for each completed data packet sent
	Green light blinking	Blinks once for each completed data packet received

Pager Light Introduction

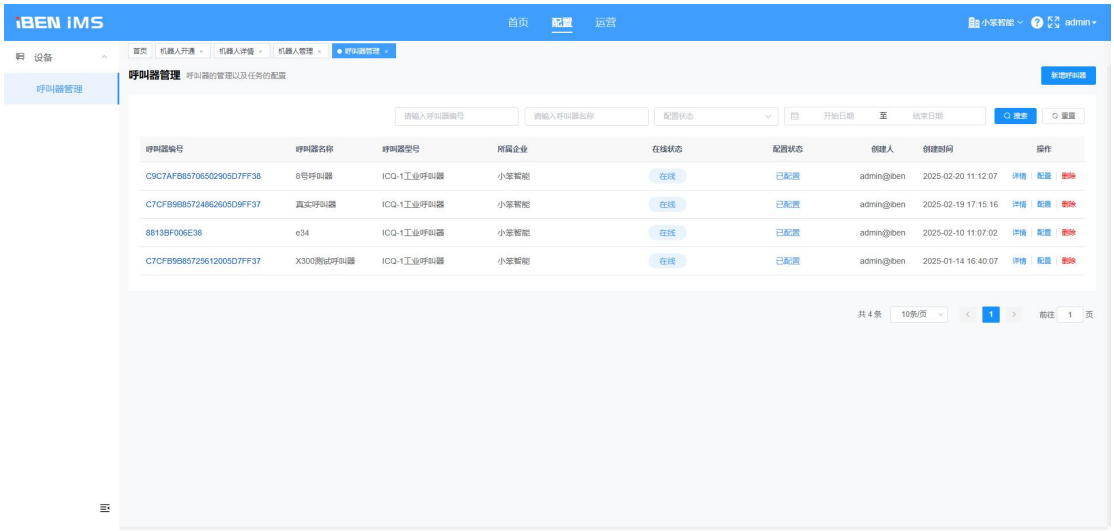
The following is the light introduction of the multi-color indicator light:

Category	Light Introduction
Indicator Light	pager status light. The traditional solution usually has 3 lights: Power indicator light: Lit up when connected (green when connected, not lit when not connected). Network connection status light: Green (connected) / Yellow (not connected). Fault light: Yellow (abnormal) / Green (no fault).
Working Status Light	**No call task:** Not lit **Press to call task:** Green slow blinking (the platform receives the

	call task and successfully creates the task) **Call successful** (after the platform responds, the task becomes in execution): Green steady on **Yellow single blink:** Call failed, no feedback from the platform; **Yellow double blink:** Robot received task failed (task scheduling abnormal, robot task execution abnormal: blocked and unable to reach, robot failure)
Buttons	● 2 buttons: 1 green and 1 red. ● Each button supports 2 tasks triggering methods: ○ Method 1: Single press trigger Method 2: Long press for 3s trigger ● Button definition: ○ Green button: ■ Single press: Execute call-to-point task (default function, can be modified in the cloud platform configuration page). ■ Long press for 3s: Custom task (configured in the cloud platform configuration page). ○ Red button: ■ Single press: Custom task (configured in the cloud platform configuration page) ■ Long press for 3s: Cancel current call task (default function, can

	be modified in the cloud platform configuration page)

2.3.2.3 Pager List



Operation button area: The "Add pager" button in the upper right corner can be used to add a new pager.

Filter area:

The search conditions are in an "AND" relationship, and all search boxes support fuzzy query.

There are input boxes for pager number and pager name, a drop-down menu for configuration status, and a start date to end date selection box.

Click the "Search" button to filter pagers according to the set conditions, and click "Reset" to clear the filter conditions.

List Information in Display Area:

- Pager Number: The unique identifier of each pager, such as "C9C7AFBB85706502905D7FF38". The pager number is usually on the outer packaging of the pager. The pager number will be obtained after the pager connects to the network and the iMS cloud platform.
- Pager Name: Such as "No.8 pager" and "Real pager".
- Pager Model: Such as "ICQ-1 Industrial pager".
- Affiliated Enterprise: The enterprise to which the pager belongs.
- Online Status: Online status (Online, Offline).
- Configuration Status: Function configuration (Configured, Unconfigured).

ⓘ Configuration rule judgment:

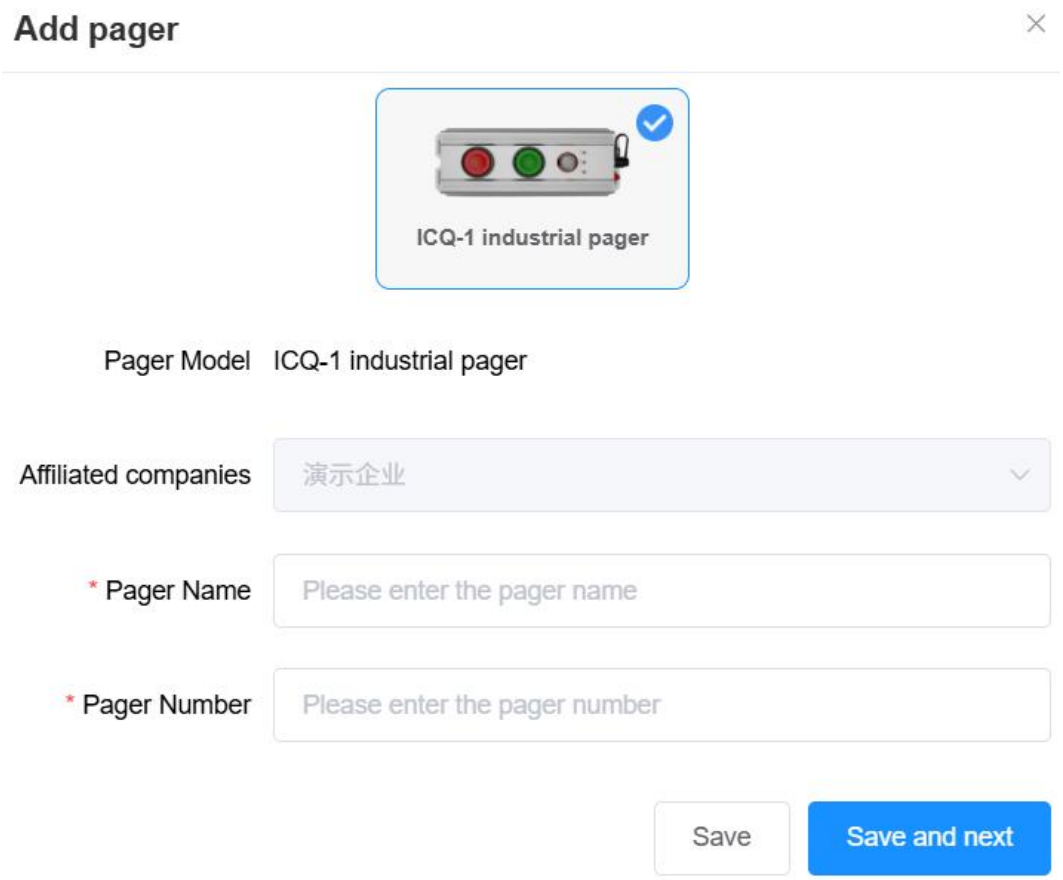
Unconfigured: If all buttons are not configured, it is displayed as Unconfigured. The Unconfigured button is highlighted and can be clicked.

Configured: If any of the N buttons is configured with a function, it is displayed as configured.

- Creator: Such as "admin@iben".
- Creation Time: Records the specific time when the pager was created, such as "2025-02-20 11:12:07".
- Operation: Includes "Details", "Configure", and "Delete" buttons. Click "Details" to view the detailed information of the pager; "Configure" to configure the pager's tasks, etc.; "Delete" to remove the pager.

ⓘ Delete: To delete the pager function, the task must be completed or canceled. If the requirements are not met (e.g., the task is in execution), a prompt will appear: "There are pending tasks initiated by this pager. Please wait for the task to be completed or cancel the task before deleting.

2.3.2.4 Add Pager



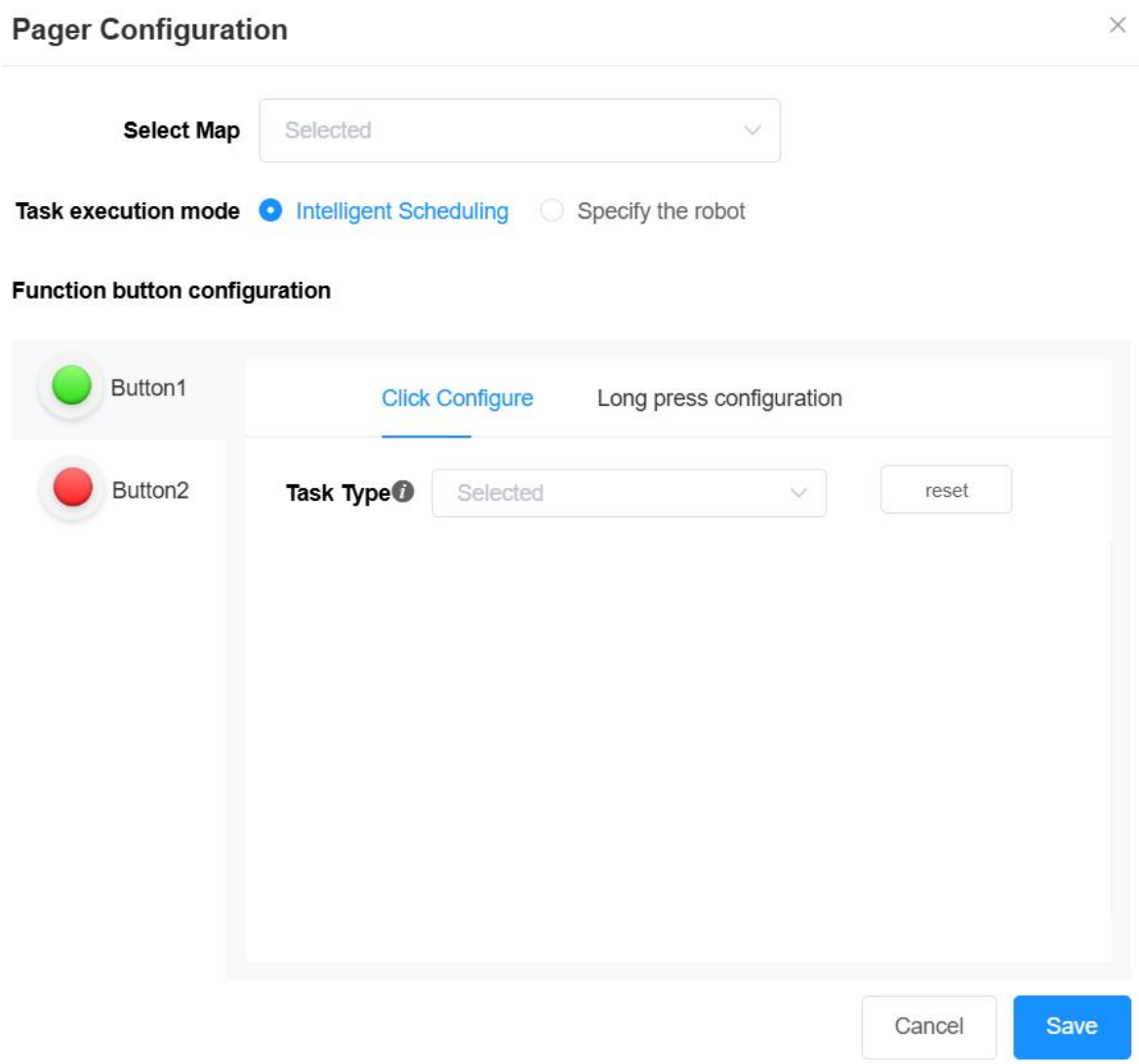
The pop-up window for adding a pager has the following function descriptions:

- Pager Display: The picture and model of the pager are displayed above to allow the user to confirm the type of pager.
- Affiliated Enterprise: The currently logged-in enterprise is displayed by default.
- Pager Name: Mandatory item, the user needs to enter the name of the pager to

identify it.

- Pager Number: Mandatory item, the user can manually enter the number or click "Select pager" to confirm the number.
- Operation Buttons: The "Save" button is used to save the pager information; the "Save and Next" button not only saves the information but also guide the user to the subsequent configuration process.

2.3.2.5 Configure Pager



1. Task Execution Mode

- Intelligent Scheduling: This is an option for task execution mode. In this mode, the system will automatically assign tasks to suitable robots according to certain rules.
- Designated Robot: The robot number "E6579ABCF9FC6A9E1" is selected in the drop-down box on the right, indicating that the task is assigned to a specific robot for execution.

ⓘ Intelligent Scheduling:

The intelligent scheduling strategy of V1.0.0 is as follows:

(1) Priority is given to robots with running status = Idle to execute tasks.

(2) When all robots are idle, the robot with the highest battery level is selected to execute the task.

(3) When the robots have the same battery level, the closest robot is selected to execute the task.

2. Function Button Configuration Area

Button Display: Displays "No.1 Button" (green) and "No.2 Button " (red).

Operation Configuration: Each button supports "Single Press Configuration" and "Long Press Configuration".

Task Type: The drop-down menu allows selecting task types, which are: Point-to-Point Task, Custom Task, and Cancel Task.

Point-to-Point Task: Obtains all waypoints of the robot and calls it to reach a certain waypoint.

Single Press Task: Obtains the task list (task list number, name) of the robot.

Custom Task Group: Custom task groups edited on the cloud platform.

Cancel Task: Cancel the task.

- Select Custom Task Group: Lists multiple task options, such as "Test Waypoint 1" and "Multi-Point Patrol", from which the specific task triggered by a single press of the button can be selected.
- Reset: Resets the task selection of the current button. After resetting, the button becomes unconfigured and can be saved. It can be used to disable the function of the button.

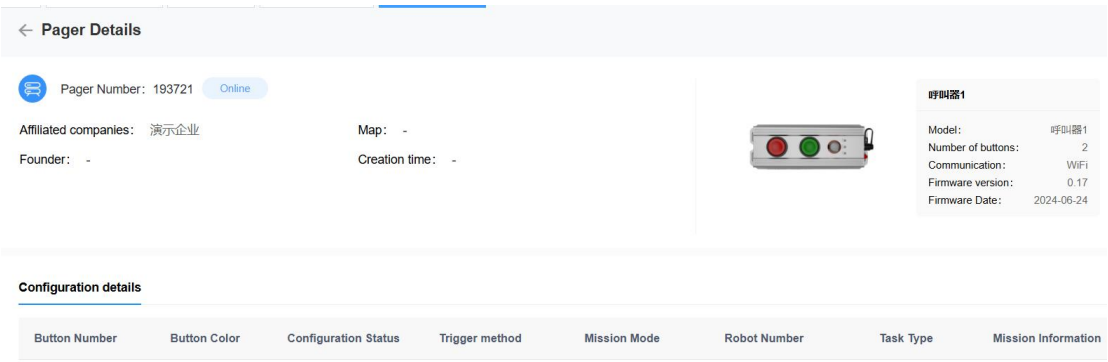
3. Operation Button Area

- Cancel: Clicking this button cancels the configuration operation without saving the settings.
- Save: Clicking this button saves the configuration settings of the pager.

Note: The overall task logic of the pager is an exclusive mode based on the pager. When a task is being executed, the multi-color indicator light is in the green steady-on mode. During the entire task process, no new tasks can be initiated. Only when the task ends and the multi-color indicator light goes out can the next task be initiated by pressing the button again.

During the task, the long-press button cancel function can be used.

2.3.2.6 Pager Details



1. Basic Information Area

- Pager Number: Displays the unique number of the pager "C9C7AFBB85706502905D7FF38", which is used to identify the pager in the system.
- Online Status: Displays "Online" or "Offline", indicating the current network connection status of the pager.
- Affiliated Enterprise: Clearly indicates the enterprise to which the pager belongs.
- Creator: Displays the user who created the pager, "admin@iben".
- Creation Time: Records the specific time when the pager was created, "2025-02-20 11:12:07".
- Affiliated Map: Currently displayed as "-", which may be used to associate the map information where the pager is located (not filled in).

2. Pager Parameter in Display Area

- Pager Name: Such as "No.8 pager".

- Model: "ICQ-1 Industrial pager".
- Number of Buttons: Displays the number of buttons.
- Communication Method: Uses "WiFi" or "Lora".
- Firmware Version: The version number is "0.19".
- Firmware Date: "2024-09-07".

3. Configuration Details Area

Displays the configuration information of each button of the pager in the form of a table:

- Button Number: Such as "No.1 key", "No.2 key".
- Button Color: Displays the color of the button, such as green, red.
- Configuration Status: Includes "Configured", "Unconfigured", and "Invalid". When the waypoint or task bound to the pager button is invalid, the configuration of the button is "Invalid" and needs to be reconfigured for use.
- Trigger Method: Includes "Single Press" and "Long Press".
- Task Mode: All are "Designated Robot".
- Robot Number: The associated robot number "ABC12390865008".
- Task Type: Such as "Custom Task", "Cancel Task".
- Task Information: Displays the specific content of the task, such as "New Task 1 to 3", "New Task 2 to 4". You can click to view the configured task information, which is displayed in a pop-up window.

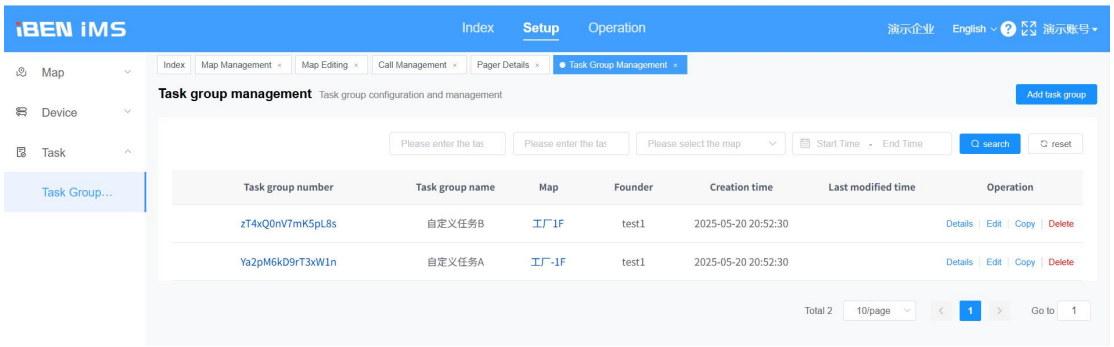
2.3.3 Task

2.3.3.1 Task Group Management

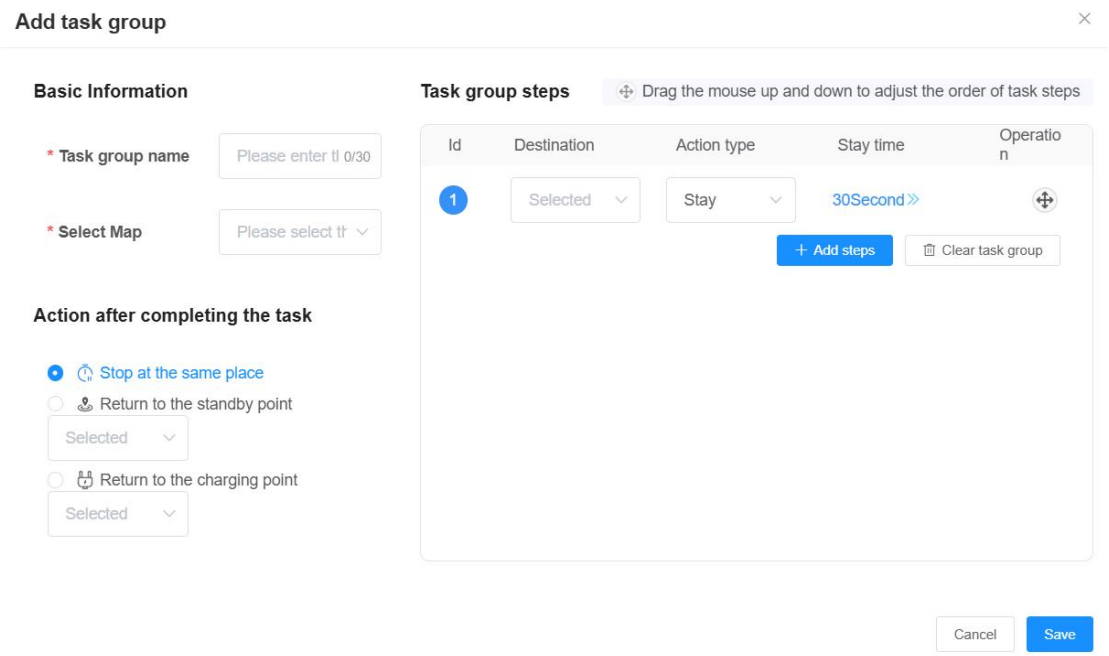
Task group management is used to configure and manage task groups, supporting the creating, editing, copying, and deleting of task groups. It can define task steps, destinations, action types, and stay time to meet diverse task scheduling needs.

Task Group List

Task Group List: Displays task group number, name, affiliated map, creator, creation time, and last modification time. It supports searching for task groups by task group number, name, affiliated map, and time range. The operation column includes buttons for details, editing, copying, and deleting.



New task group pop-up window:



Basic Information:

- Task Group Name: Mandatory, maximum 30 characters.
- Select Map: Mandatory, select the map associated with the task group.
- Action After Completing Task: Optional "Stay in Place" or "Return to Charging Point" (if "Return to Charging Point" is selected, the charging point needs to be further selected).

Task Group Steps:

- Steps can be added. Each step requires selecting a destination, action type (such as Stay in place), and setting the duration of stay.
- Supports dragging up and down to adjust the order of steps, clicking "-" to delete a step, "+ Add Step" to add a new step, and "Clear Task Group" to clear all steps.

Duration of Stay Setting Pop-up Window:

- Manual Confirmation to Leave: The robot waits until manual confirmation to continue the task.
- Custom Duration of Stay: Set the specific seconds (adjustable via "-", "+" or by dragging the slider). The robot automatically continues the task after the setting time.

2.4 Operation

The operation menu is the first-level menu of the entire system, which manages the overall operation of the entire enterprise.

2.4.1 Monitor Task

The task module is mainly used for monitoring and managing the tasks status.

Definition of Task:

A task refers to a complete handling work unit that a robot needs to complete.

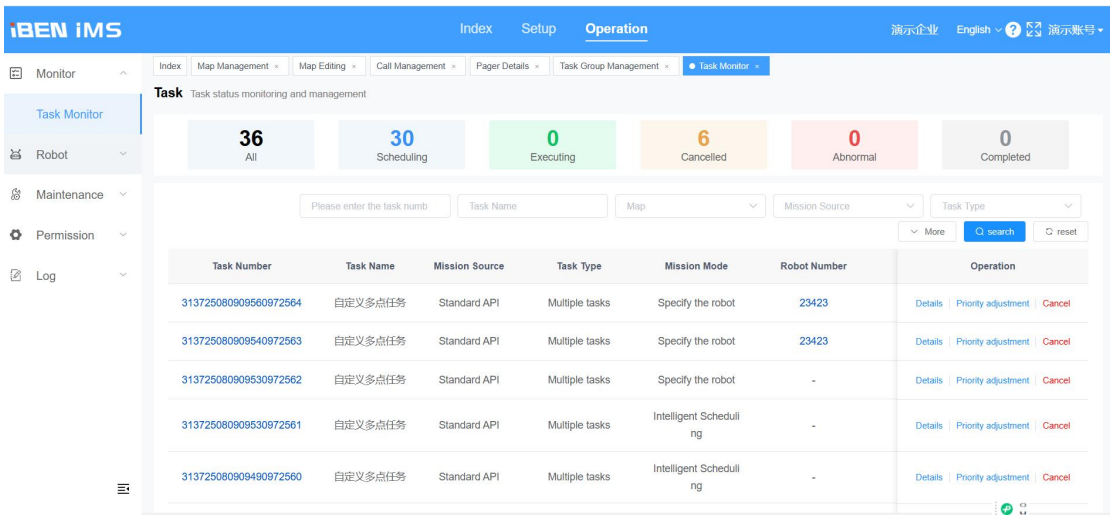
It includes the entire process of transporting materials or products from the starting point to the target location.

Example of Task:

In industry, tasks are usually transferred according to the workflow. For example, the production of a Tesla requires a complete production process, which includes

installing tires at point A, installing glass at point B, installing the car body at point C, and inspecting and leaving the warehouse at point D. This entire process is collectively referred to as a task.

Therefore, a task is also usually called a process, and a task template is usually called a process template.



2.4.1.1 Task Status Overview

The task module is mainly used for monitoring and managing the tasks status. At the top of the page, you can view the total number of tasks and the number of tasks in different statuses. The task statuses include All, To Be Issued, In Execution, Canceled, Abnormal, and Completed, which is convenient for users to quickly grasp the overall situation of tasks.

2.4.1.2 Task Search and Filter

Task search and filter functions are provided. Users can search precisely and filter by

entering the task number, selecting the task status (such as In Execution), task source, task type, and setting the start date to end date to quickly locate the required tasks.

2.4.1.3 Task List Display

The data of the task list is updated every 3 seconds.

The list displays key information of the task, including task number, task name, task source, task type, task mode, robot number, status, description, and creation time.

Click "Details" to view more detailed information about the task.

ⓘ (1) When a task is abnormal, abnormal information will be displayed and can be quickly seen in the list.

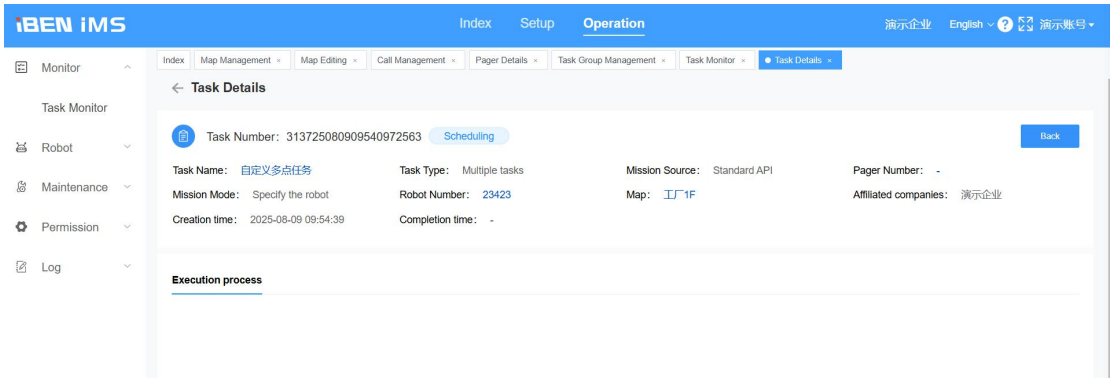
(2) When a task is in execution, the current stage of the task needs to be displayed.

(3) For tasks To Be Executed, Canceled, and Completed, no relevant display is required.

2.4.1.4 Task Operation

For tasks in execution, you can click the "Cancel" to end the task. Through these operations, the effective management of the entire life cycle of the task is realized.

2.4.1.5 Task Details Page



Task Basic Information Area

- Task Number: "1892518295849009152", which is the unique identifier of the task.
- Task Status: Displays all statuses of the task, indicating the execution result of the task.
- Task Name: For example: "New Task 2 to 4".
- Task Type: "Custom Task".
- Task Source: "pager", and the pager number is "C7CFB98B572482605D9FF37".
- Task Mode: "Designated Robot".
- Robot Number: "ABC12390865008".
- Affiliated Map: Displays specific map information (-).
- Affiliated Enterprise: "iBEN Robot".
- Creation Time: "2025-02-20 18:14:58".
- Completion Time: "2025-02-21 13:55:04".
- Starting Point and End Point: Display specific waypoint information. The

starting point of a point-to-point task is empty, and the end point is the task point.

- Operation Button: The "Return" button in the upper right corner can return to the previous page.

Execution Process Area

The execution process is a record of the information about the robot's execution of the task. It includes all the life cycles of a task.

It includes the basic information, the execution status, the execution time, and possible abnormal situations of the task.

Task Execution Process Record = Task Execution Process Record of Cloud Platform + Task Execution Process Record of Robot + Interaction Process Record between Cloud Platform and Third-Party System.

The task execution steps are displayed in the form of a timeline, including "Task Started" (2025-02-21 13:54:15), "Assigned Robot" (2025-02-21 13:54:15), two "Arrived" (at 2025-02-21 13:54:35 and 2025-02-21 13:54:58 respectively), and "Task Completed" (2025-02-21 13:55:04), and the relevant waypoint numbers are recorded.

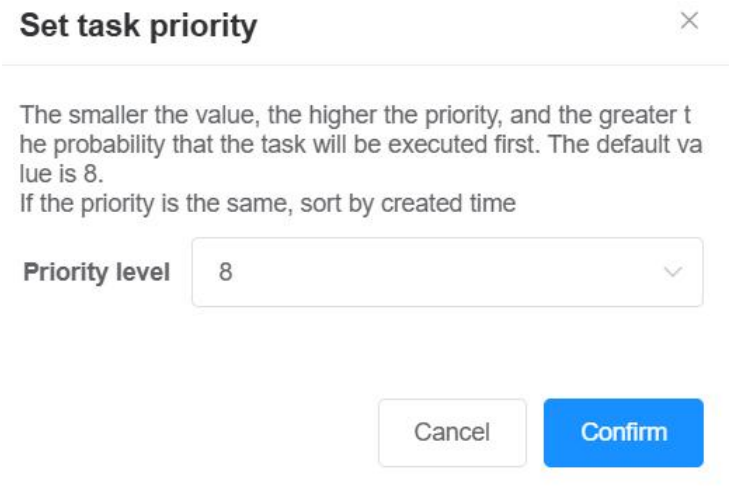
2.4.1.6 Task Priority

After a task is created, it will automatically enter the scheduling queue, with the default status of "To Be Issued". When the robot is scheduled, the task will be processed and enter the "In Execution" status.

During this process, the priority of the task can be adjusted to ensure that high-priority tasks can be issued first.

The smaller the value, the higher the priority, and the higher the probability that the task will be executed first. The default value is 8.

When the priorities are the same, the tasks are sorted by the creation time.



2.4.2 Robot

Robot Registration: After leaving the factory, robots will be registered on the cloud platform, and robot information will be managed through the cloud platform.

Robot Activation: Robots that have not activated cloud services cannot use cloud platform services, including map synchronization and participation in scheduling.

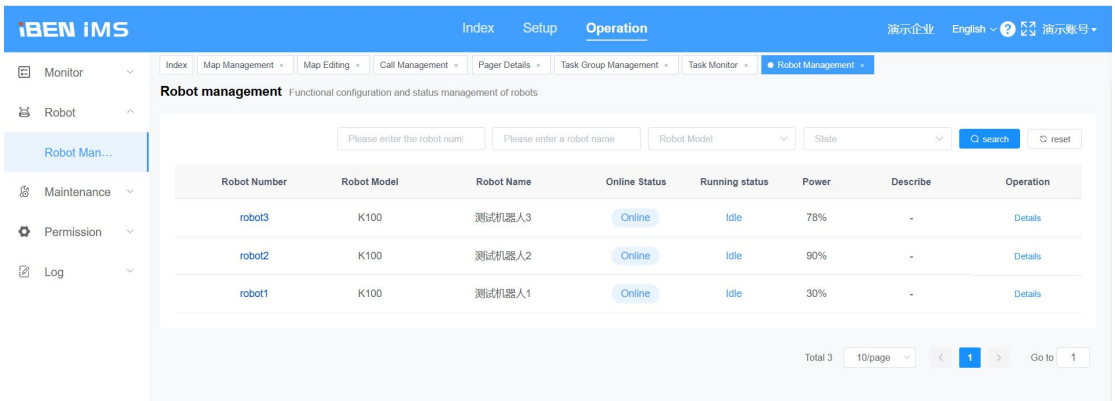
After activating, they can be used normally.

The robot activation function is only available to the super administrator of iBEN Robot, who can activate and assign robots to customers.

2.4.2.1 Robot Activation

Manage the activation and assignment of iMS cloud platform services.

1. Activated Robots



Navigation and Filter Area

Navigation Tabs: Include "Activated Robots" and "Inactivated Robots". Click the corresponding tab to switch and view the robot lists in different statuses.

Filter Bar: There are multiple filter condition input boxes and drop-down menus. Users can quickly and accurately filter out the required robot information by entering the "Robot Number" and "Enterprise Name", selecting the "Status" and "Robot Model", and setting the "Start Date" to "End Date". The "Search" button is used to execute the filtering operation, and the "Reset" button can clear the

entered filter conditions and restore the initial state.

List Information in Display Area

The list displays detailed information of activated robots, with each row corresponding to one robot. The specific information includes:

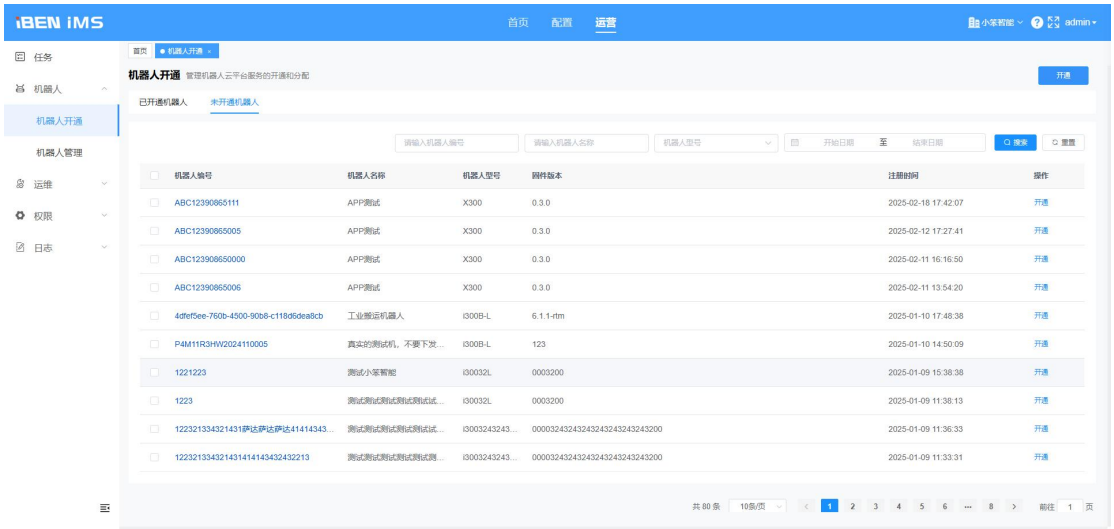
- Robot Number: The unique identifier of each robot, such as "ABC12390865008".
- Robot Name: The name of the robot, such as "APP Test" and "Industrial Handling Robot".
- Robot Model: Indicates the model of the robot, such as "X300" and "I300B-L".
- Affiliated Enterprise: Displays the enterprise to which the robot belongs, which is "iBEN Robot" on this page.
- Status: Shows the online or offline status of the robot, such as "Online" and "Offline".
- Activation Time: Records the specific time when the robot activates the cloud platform service, such as "2025-02-19 10:11:32".
- Service Duration: Counts the service duration of the robot since activation, such as "2 days" and "41 days".

Operation Area

At the far right of each row of robot information, there is an "Unbind" button. Clicking this button can unbind the corresponding robot from the cloud platform

service, so as to meet the enterprise's needs for robot service management.

2. Inactivated Robots



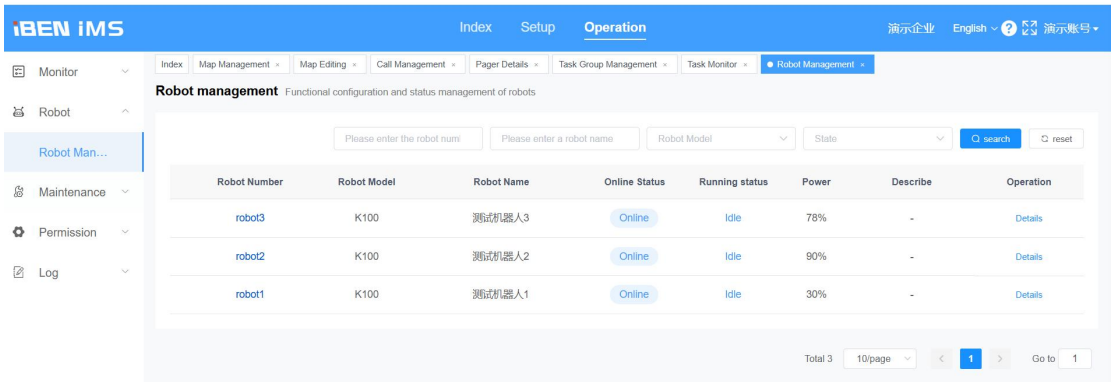
List Information in Display Area

- Robot Number: The unique identification of each robot, such as "ABC12390865111", which is used to accurately distinguish and locate different robots in the system.
- Robot Name: The name given to the robot for easy identification and management, such as "APP Test" and "Industrial Handling Robot", which is convenient for quickly understanding the purpose or project of the robot.
- Robot Model: Indicates the specification and type of the robot, such as "X300" and "i300B-L". Different models have different functions and performance characteristics, which is convenient for users to distinguish.
- Firmware Version: Displays the firmware version used by the robot, such as "0.3.0" and "6.1.1-tfm", which helps users understand the software status of the robot and determine whether firmware upgrade is needed.

- Registration Time: Records the specific time when the robot completes registration in the system, in the format of "Year-Month-Day Hour: Minute: Second", such as "2025-02-18 17:42:07", which is used to trace the registration history and service duration.
- Operation Button: For inactivated robots, the "Activate" button is displayed. Clicking it can activate the cloud platform service for a single robot; for activated robots, the "Unbind" button is displayed. Clicking it can unbind the robot from the cloud platform service.

2.4.2.2 Robot Management

Function configuration and status management of robots.



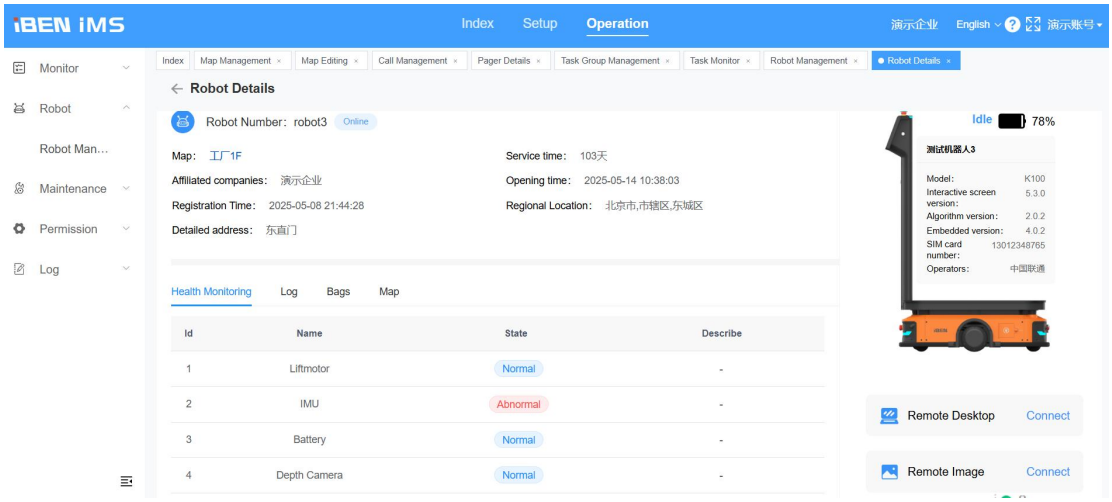
The list is updated every 3 seconds.

List Information in Display Area

- Robot Number: The unique identifier of the robot, such as "ABC12390865008".
- Robot Model: Displays the specific model, such as "X300" and "I300B-L".
- Robot Name: Such as "APP Test" and "Industrial Handling Robot".
- Affiliated Enterprise: All are "iBEN Robot".

- Status: Displays "Online" or "Offline".
- Battery: Displays the current battery percentage of the robot, such as "40%" and "100%".
- Operation: Click "Details" to view the detailed information of the robot.

2.4.2.3 Robot Details



1. Basic Information Area

- Robot Number: Displays the unique number "ABC12390865008" for identifying the robot.
- Online Status: Displays "Online", indicating the current network connection status of the robot.
- Affiliated Map: Currently displayed as "-", which may be used to associate the map information where the robot is located.
- Service Duration: Displays "2 days", recording the duration of the robot has been put into use.

- Affiliated Enterprise: Marked as "iBEN Robot" to clearly indicate the affiliation.
- Activation Time: Recorded as "2025-02-19 10:11:32".
- Registration Time: "2025-02-19 10:04:53".
- Region Location: Displays "Beijing City, Chaoyang District", indicating the region where the robot is located.
- Detailed Address: Specific address or related information.

2. Right Information in Display Area

- Status and Battery: Displays the robot status as "Idle" and the battery level as "38%".
- Name and Model: The name is "APP Test" and the model is "X300".
- Software and Firmware Versions: Software version "1.0.0", firmware version "0.3.0".
- SIM Card Number and Operator: If configured, the SIM card number and operator information will be displayed.

3. Lower Menu Area

- Health Monitor: Used to display health-related information of the robot, such as hardware status.
- Log: Allows viewing and obtaining various log records during the robot's operation.
- Bag: Allows viewing and obtaining bag log records during the robot's

operation.

- Map: May be used to view the map of the robot, which can be viewed and downloaded.

Definition of Robot Operating Status

	状态（中文）	颜色	状态（英文）	规则说明
1	空闲	空闲	Idle	机器人没有执行任务
2	充电中	充电中	Charging	机器人充电中
3	工作中	工作中	Working	机器人执行任务中
4	异常	异常	Error	机器人软件硬件运行异常
5	急停	急停	EmergencyStop	机器人急停中

1. Robot Health Monitoring

Health Monitoring Log Bags Map

Id	Name	State	Describe
1	Liftmotor	Normal	-
2	IMU	Abnormal	-
3	Battery	Normal	-
4	Depth Camera	Normal	-
5	Odometer	Normal	-
6	laser	Normal	-

Robot health monitoring displays the operating status of each component of the robot:

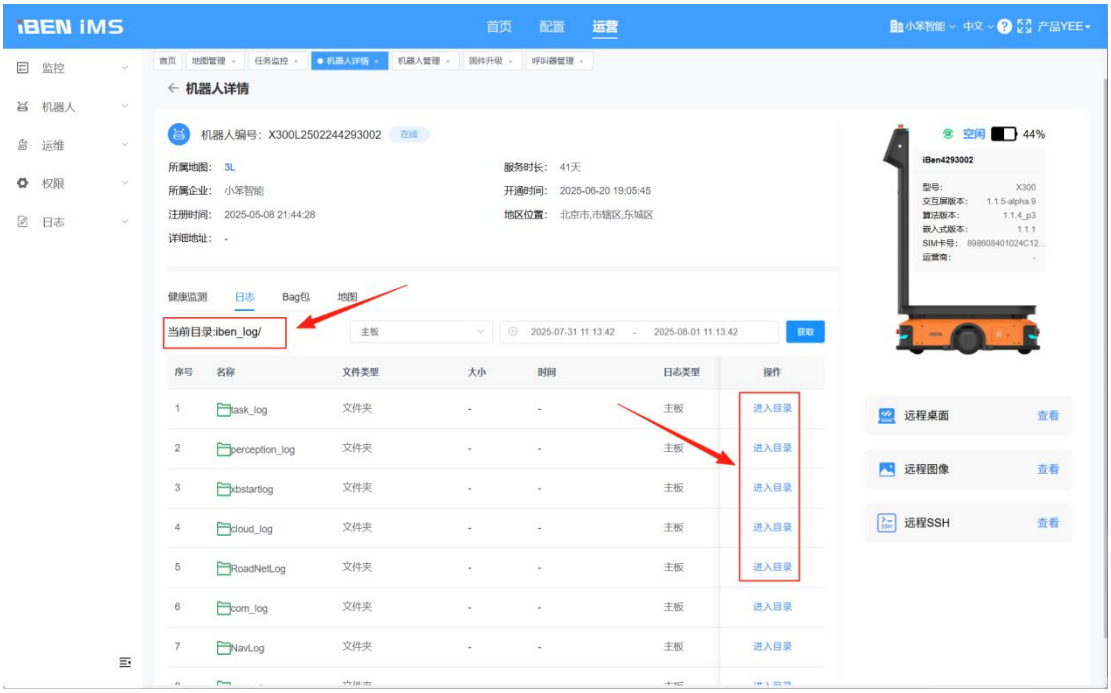
- Motor: Number 1, status "Operating Normally".

- IMU (Inertial Measurement Unit): Number 2, status "Operating Normally".
- Battery: Number 3, status "Operating Normally".
- Depth Camera: Number 4, status "Operating Abnormally", error code 20085.
- Odometer: Number 5, status "Operating Normally".
- Laser: Number 6, status "Operating Normally".

2. Robot Log

Remotely obtain robot logs to analyze problems, reduce operation and maintenance costs, and improve operation and maintenance efficiency.

The hierarchical display of the robot's mainboard log folder allows retrieving the log files freely and accurately.



Filter Area: You can filter log files by selecting the log "Type" (such as interactive screen software), setting the time range (such as "2024-09-28 15:00 ~ 16:00"), and clicking the "Obtain" button.

Log List Area: Displays log file information in the form of a table:

- Serial Number: Sequence number of the log file.
- File Name: Such as "test-20220323-131711.log".
- Size: All is "152 KB".
- Time: Records the log generated time, such as "March 15, 2025 07:46:20".
- Log Type: Includes "Interactive Screen", "Mainboard", etc.
- Status: Displays the log acquisition status, such as "Not Obtained", "Obtaining", "Obtained Successfully", "Obtaining Failed".
- Operation: Includes "Obtain" and "Download to Local" buttons, which can be used to obtain log files and download them to the local computer for saving.

3. Robot Bag Package

Remotely obtain robot bag packages to analyze problems, reduce operation and maintenance costs, and improve operation and maintenance efficiency.

After the robot is turned on, it will start recording the bag package. The bag package can store data for about 3 hours.

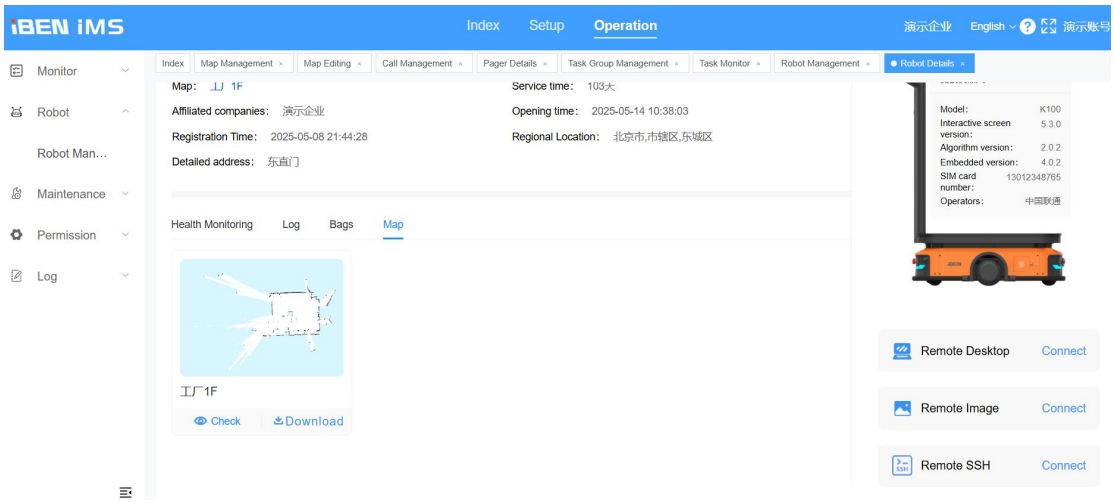
Filter Area: There is a red prompt in the upper right corner: "Only dates within the last 7 days can be selected, and bags can be checked by the hour". Users can set the time range in the date selection box, such as "2024-09-28 15:00 ~ 16:00", and then click the "Obtain" button to filter the bag package files in the corresponding time period.

List Information in Display Area: Displays information related to bag package files in the form of a table:

- Serial Number: Sequence number of the file.
- File Name: Such as "bag-20250315-074620-6.zip".
- Size: Displays the file size, such as "230 MB", "225 MB", etc.
- Time: Records the file generated time, such as "March 15, 2025 07:46:20".
- Status: Displays the acquisition status of the file, including "Not Obtained", "Obtaining", "Obtained Successfully", "Obtaining Failed".
- Operation: Includes "Obtain" and "Download to Local" buttons, which are used to obtain the bag package file and download it to the local computer for saving.

4. Robot Map

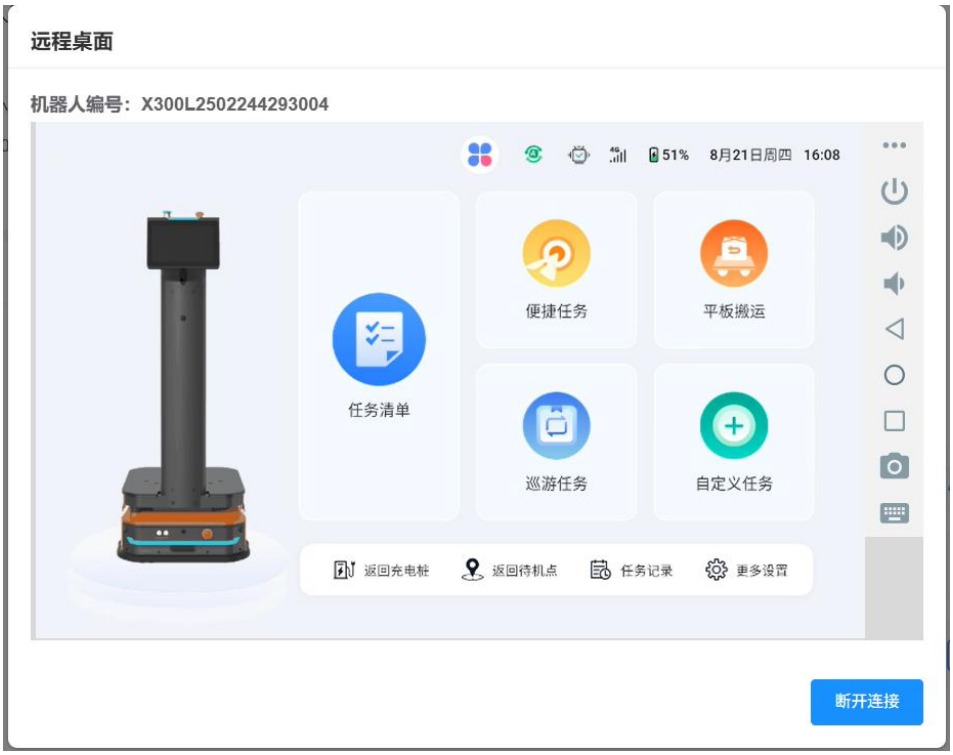
View the map currently used by the robot, which can be viewed and downloaded.



- Map Preview: A map thumbnail is displayed to allow a preliminary view of the map appearance.
- Operation Buttons: The "View" button is used to open the map for detailed viewing; the "Download" button can download the map to the local computer.

5. Remote Desktop

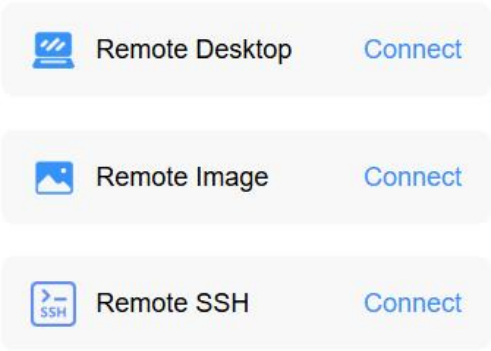
After activating the remote desktop function permission, you can connect to the interactive screen of on-site robot to solve problems.



6. Remote Image

Remote images can be controlled through permissions. After opening, you can directly access the middle camera on the robot, view the real-time on-site screen, and assist in troubleshooting.

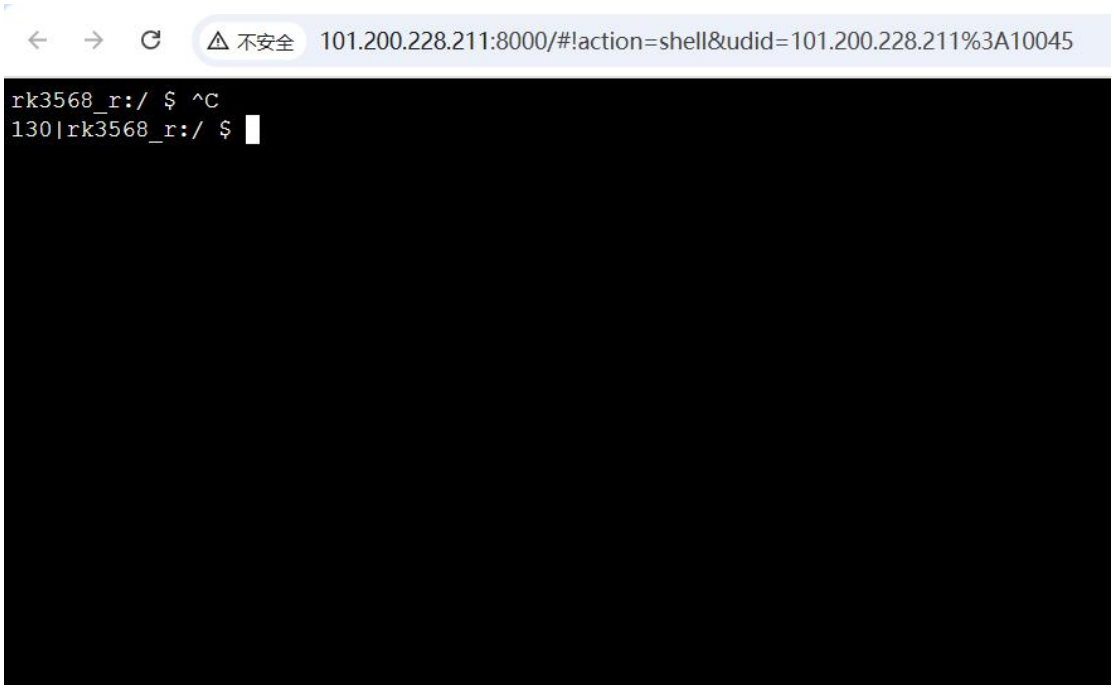
Idle
78%
测试机器人3
Model: K100
Interactive screen version: 5.3.0
Algorithm version: 2.0.2
Embedded version: 4.0.2
SIM card number: 13012348765
Operators: 中国联通


Remote Desktop
Connect
Remote Image
Connect
Remote SSH
Connect

远程图像
2025-07-21 18:32:54
187.58KB/s
WASM
标准
断开连接

7. Remote SSH Function

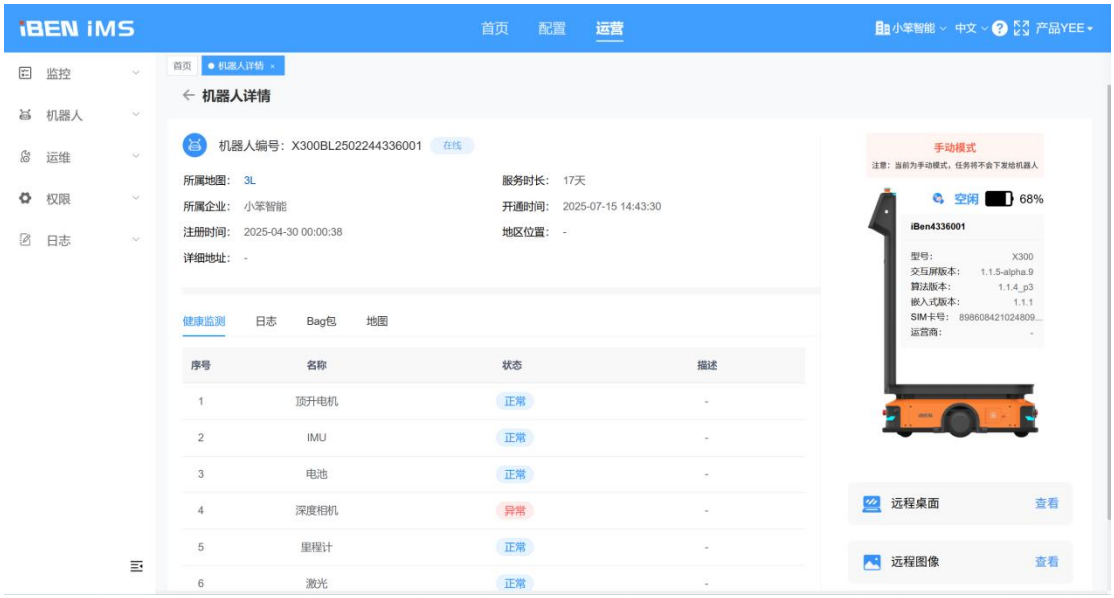
Remote SSH allows users to establish an encrypted connection between iMS and the robot, so as to perform command-line operations, file transfer, log viewing and other management work on the remote device.



8. Robot Mode Switch

The robot supports switching between manual mode and automatic mode. The current status of the robot can also be seen on the robot details page of the cloud platform.

When the robot is in manual mode, no tasks will be issued. At this time, maintenance operations can be performed on the robot.



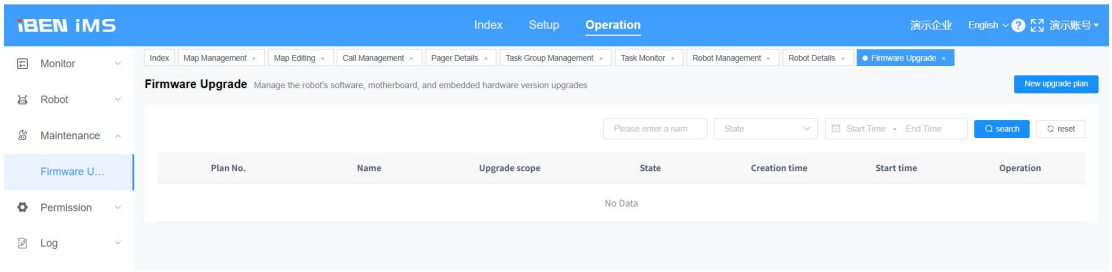
2.4.3 Operation and Maintenance

The operation and maintenance module includes relevant tools required for technical support to reduce operation and maintenance costs and improve efficiency.

2.4.3.1 Firmware Upgrade

The firmware upgrade page is used to manage the upgrade operations of the robot's software, mainboard, embedded hardware and other versions. It can create, manage and monitor upgrade plans to ensure that the robot firmware remains up-to-date and improve performance and stability.

Firmware Upgrade List



（1）Display of Upgrade Plan List

The page displays firmware upgrade plans in the form of a list, including the following information:

Plan Number: A unique identifier for each upgrade plan, e.g., "189".

Name: The name of the upgrade plan.

Upgrade Scope: Indicates the types of firmware involved in the upgrade, e.g., "Interactive Screen Firmware, Mainboard Firmware, Embedded Firmware".

Status: The current status of the upgrade plan, including "To Be Issued", "Canceled", "Being Pushed", etc.

Creation Time: Records the specific time when the upgrade plan was created, e.g., "2025-02-21 13:39:06".

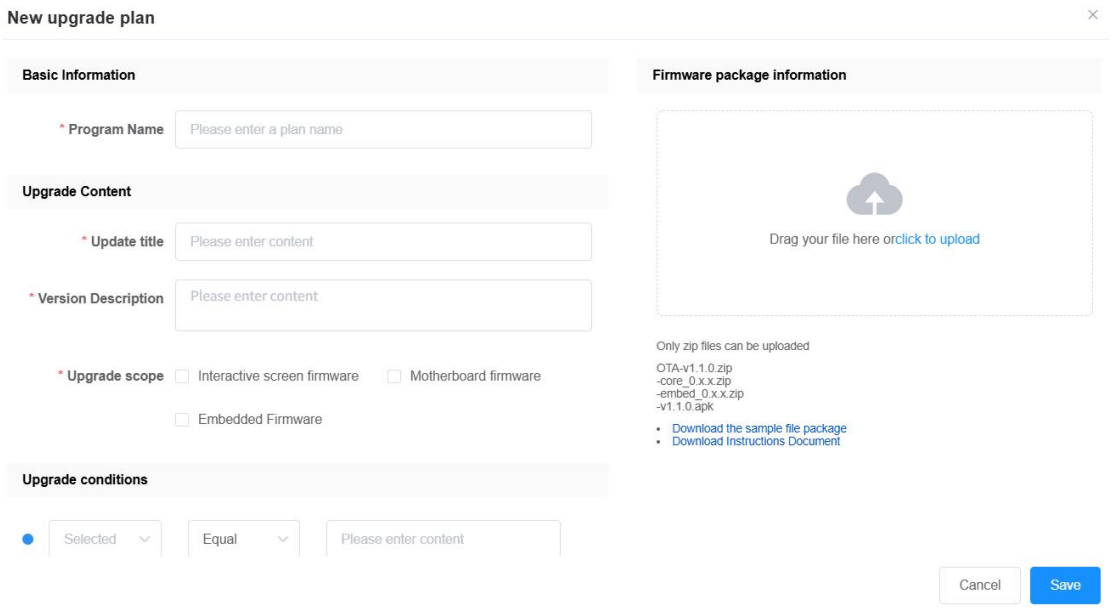
Start Time: Displays the time when the upgrade plan is scheduled to start execution; it is empty if the upgrade has not yet begun.

Operations: Provides operation options such as "Details", "Start", and "Cancel".

Button Availability by Status:

- To Be Issued: "Details", "Start", and "Cancel" are available.
- Being Pushed: "Details" and "Cancel" are available.
- Canceled: Only "Details" is available.

Adding a New Upgrade Plan



- Basic Information
- Plan Name: Mandatory field. Users must enter a name for the upgrade plan to identify it.
- Upgrade Content

Update Title: Mandatory field. Users need to enter a title for the firmware upgrade to briefly summarize the upgrade content.

Version Description: Mandatory field. Users provide a detailed description of the firmware version update, including improvements, etc.

Upgrade Scope: Users can check options such as "Interactive Screen Firmware", "Mainboard Firmware", and "Embedded Firmware" to specify the types of firmware included in the upgrade.

- Firmware Package Information

File Upload Area: Users can drag firmware package files (only zip files are supported) into this area or click the area to upload files, providing the upgrade firmware

package.

- Upgrade Conditions

Condition Settings: By default, the condition is set to "Robot Number", but users can also select "Map Number". Users can choose logical relationships such as "Equal" and enter the corresponding robot number. Click "+ Add" to add more upgrade conditions (e.g., multiple robot numbers or other criteria) to precisely control the upgrade scope.

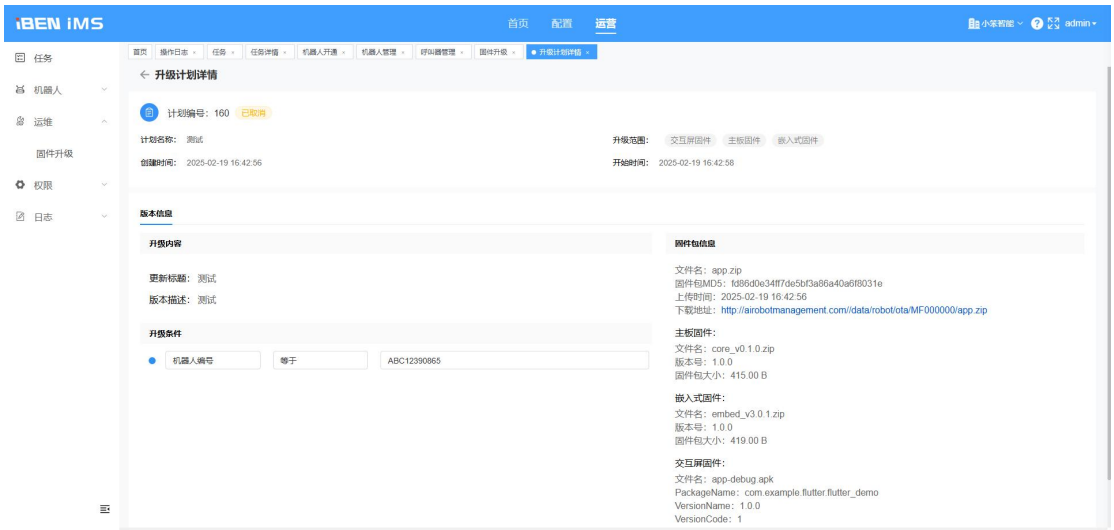
- Operation Buttons

Cancel: Clicking this button, cancels the operation of adding a new upgrade plan, closes the pop-up window, and does not save any input information.

Save: Clicking this button, saves the upgrade plan information entered by the user, creates a new upgrade plan, and displays it in the upgrade plan list on the firmware upgrade page.

ⓘ Currently, OTA firmware packages (including interactive screen firmware, embedded firmware, and mainboard firmware) are all packaged into a .zip file. After uploading the firmware package, the system automatically parses and extracts relevant information such as version numbers and MD5 values.

Upgrade Plan Details



Basic Information Area

Plan Number: Displays the unique number of the upgrade plan (e.g., "160"), used to accurately identify and distinguish different upgrade plans in the system.

Plan Status: Intuitively shows the current status of the plan (e.g., "Canceled"), helping users quickly understand the progress of the plan.

Plan Name: Displays the name of the plan.

Creation Time: Records the specific time when the plan was created, e.g., "2025-02-19 16:42:56".

Upgrade Scope: Clearly lists the types of firmware involved in the upgrade, including "Interactive Screen Firmware", "Mainboard Firmware", and "Embedded Firmware".

Start Time: Displays the time when the upgrade plan starts (e.g., "2025-02-19 16:42:58"), helping users track the start time of the upgrade.

Version Information Area

Update Title: Provides a brief summary of the firmware upgrade (e.g., "Test"),

helping users quickly understand the main content of the upgrade.

Version Description: Offers detailed information about the firmware version update (in this case, "Test"). Typically, it should include details such as feature improvements and bug fixes.

Upgrade Conditions: Explains the conditions that trigger the upgrade. In this example, the condition is "Robot Number = ABC123908655", specifying the specific robot targeted by the upgrade plan.

Firmware Package Information Area

For different types of firmware (Mainboard Firmware, Embedded Firmware, Interactive Screen Firmware, etc.), the following information is displayed respectively:

File Name: Shows the specific name of the firmware package (e.g., "core_v0.1.0.zip" for mainboard firmware), users verify the file quickly.

Version Number: Indicates the firmware version (e.g., "1.0.0"), used to compare version differences before and after the upgrade.

Firmware Package Size: Specifies the size of the firmware package (e.g., "415.00 B"), allowing users to estimate the storage space required for download and the approximate download time.

Some firmware also displays information such as "PackageName" (package name) to more accurately identify the properties and source of the firmware.

For the overall firmware package, the following information is also displayed:

Firmware Package MD5: e.g., "fd86d0e34ff7de5b3a8a6a40a6f8031e", used to verify the integrity and accuracy of the firmware package.

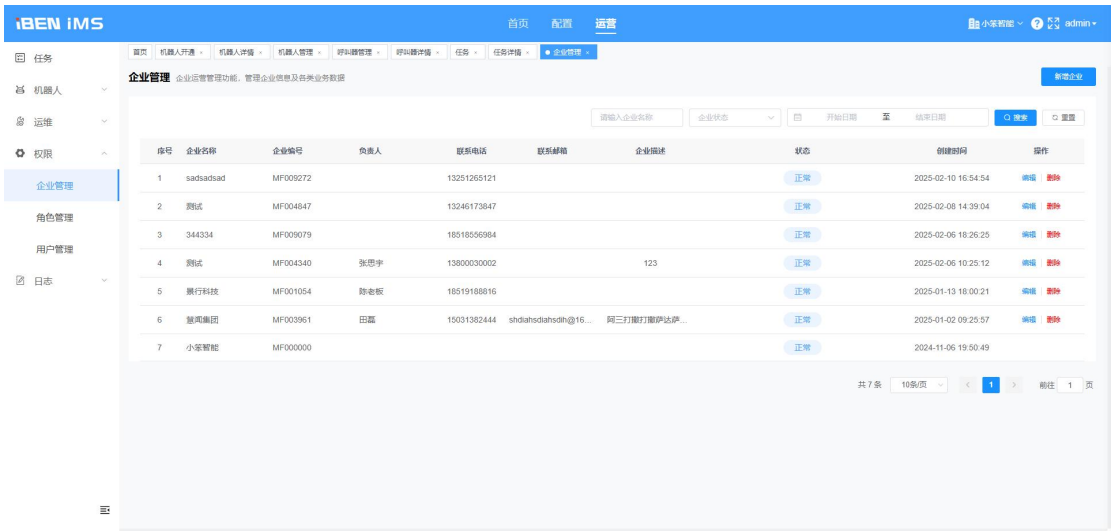
Upload Time: Records the time when the firmware package was uploaded to the system (e.g., "2025-02-19 16:42:56"), enabling traceability of the file upload timeline.

Download Address: Provides a download link for the firmware package (e.g., "http://airobotmanagement.com/data/robot/ota/MF000000/app.zip"), facilitating re-acquisition of the firmware package when needed.

2.4.4 Permission

2.4.4.1 Enterprise Management

Operation and management functions for managing enterprise information and various types of business data.



List Information in Display Area: Enterprise-related information is presented in a table format:

- Serial Number: The sequential number of the enterprise record.
- Enterprise Name: e.g., "iBEN Robot", "Test", "JX Technology", etc.
- Enterprise Number: A unique identifier for each enterprise, e.g., "MF009272", "MF004847", etc.
- Person in Charge: Records the person in charge of the enterprise, e.g., "Zhang Siyu", etc.
- Contact Phone: Displays the contact phone number of the person in charge, e.g., "13251265121", "13800030002", etc.
- Contact Email: Some enterprises have a contact email displayed.
- Enterprise Description: A brief description of the enterprise.
- Status: All displayed as "Normal".
- Creation Time: Records the specific time when the enterprise information was created, e.g., "2025-02-10 16:54:54", "2024-11-06 19:50:49", etc.
- Operations: Includes "Edit" and "Delete" buttons for modifying or deleting enterprise information.

Adding a New Enterprise

The pop-up window for adding a new enterprise has the following specific functions:

Mandatory Information:

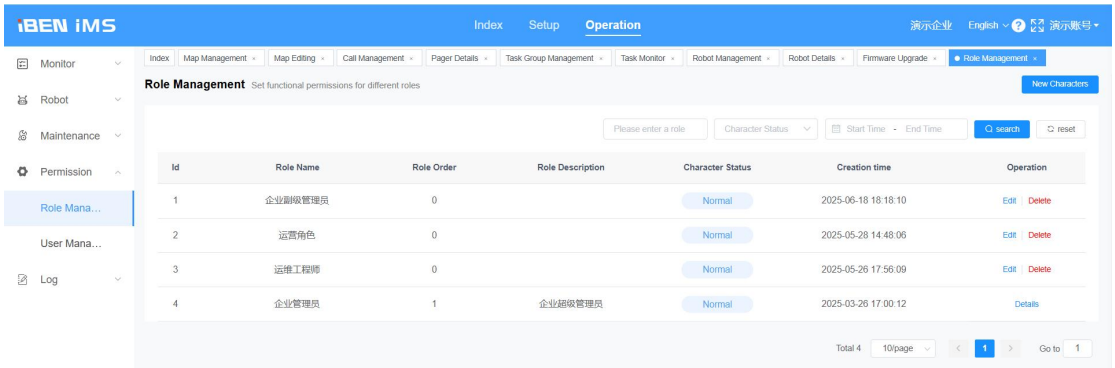
- Enterprise Name: Users must enter the name of the enterprise as its identifier.
- Contact Phone: Enter the enterprise's contact phone number for easy communication.
- Person in Charge: Enter the name of the enterprise's person in charge.
- Contact Email: Enter the enterprise's contact email address.
- Enterprise Status: Can be set to "Normal" or "Deactivated". "Normal" means the enterprise is in a state of normal operation and management; "Deactivated" restricts enterprise-related operations.
- Enterprise Description: Users can enter a brief introduction or relevant

descriptive information about the enterprise.

- Operation Buttons: “Confirm” Clicking this button submits the information of the new enterprise.
- “Cancel” Clicking this button cancels the operation and does not save the entered content.

2.4.4.2 Role Management

Configure function permissions for different roles.



Operation Button Area: The "Add Role" button in the upper right corner is used to create a new role.

List Information in Display Area:

Role-related information is displayed in a table format:

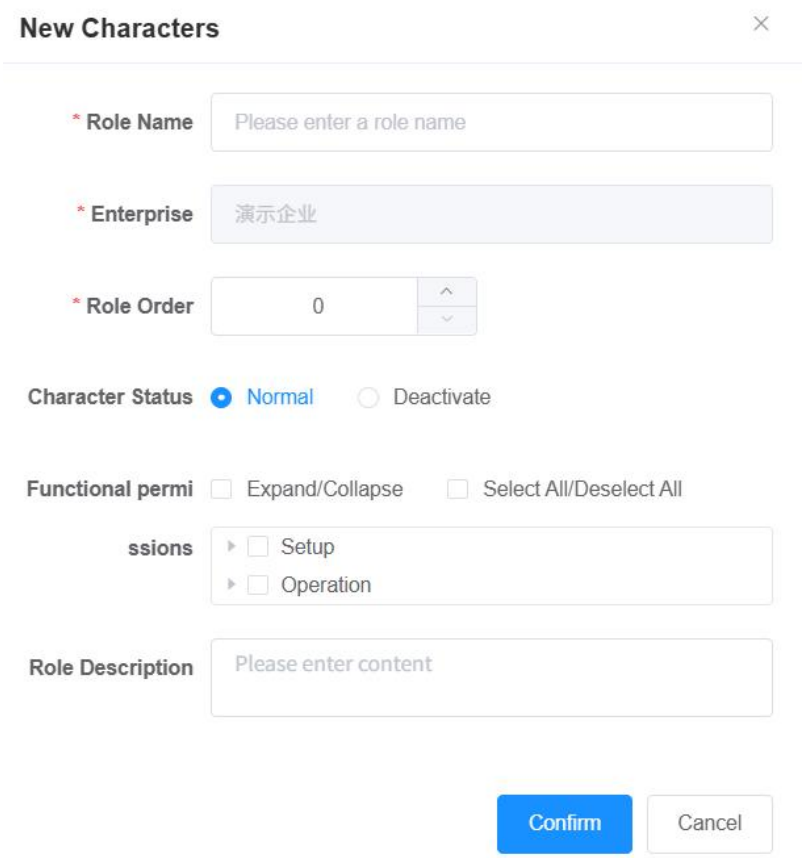
- Serial Number: The sequential number of the role record.
- Role Name: e.g., "12", "Test", "General Employee", etc.
- Display Order: Some roles have a display order value, e.g., "0", "1".
- Role Description: A brief description of the role, e.g., "Has all permissions",

"Super Administrator", etc.

- Status: All displayed as "Normal".
- Creation Time: Records the specific time when the role was created, e.g., "2024-12-28 10:41:04", "2025-02-06 16:34:18", etc.
- Operations: Includes "Edit" and "Delete" buttons for modifying or deleting role information.

Adding a New Role

ⓘ The system has 2 preset special roles: Super Administrator and Enterprise Administrator. When opening an enterprise account, users can select the permission of the Enterprise Administrator role. The permission control for the Enterprise Administrator role is configured when the system goes online, and some functions cannot be used. The Enterprise Administrator is the user with the highest permissions in the enterprise and can create sub-accounts and sub-roles under each enterprise respectively.



The pop-up window for adding a new role in the iMS system has the following function descriptions:

Mandatory Information:

- Role Name: Users must enter a name of the role to be created to identify it.
- Enterprise: Displays "iBEN Robot" by default, indicating the enterprise to which the role belongs (may not be modifiable based on existing content).
- Role Order: Sets the display order of the role, with a default value of "0", which can be adjusted via the drop-down arrow.
- Role Status: Can be set to "Normal" or "Deactivated". "Normal" means the role can be used normally; "Deactivated" restricts the use of the role.
- Function Permissions:

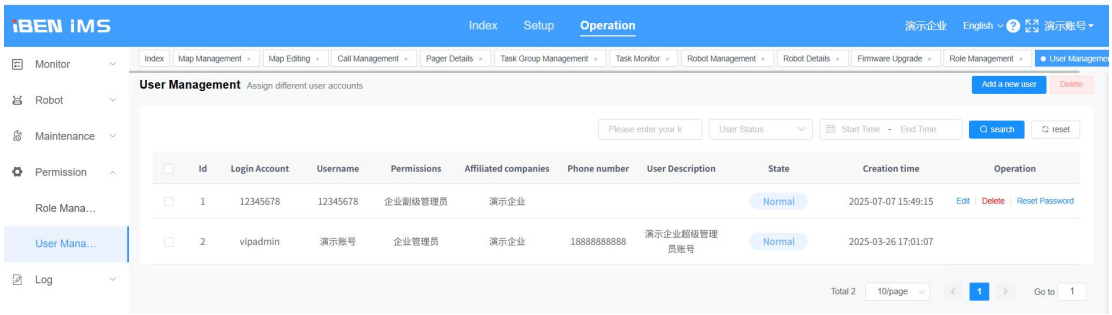
There are shortcut operation buttons for "Expand / Collapse" and "Select All / Select None".

Specific permission options include "Configuration" and "Operation". Users can check the corresponding permissions to assign operational rights to the role.

- Role Description: Users can enter descriptive information about the role to facilitate understanding of the role's responsibilities and purpose.
- Operation Buttons: “Confirm” Clicking this button saves the settings for the new role.“Cancel” Clicking this button cancels the operation and does not save the entered content.

2.4.4.3 User Management

Assign accounts to different users.



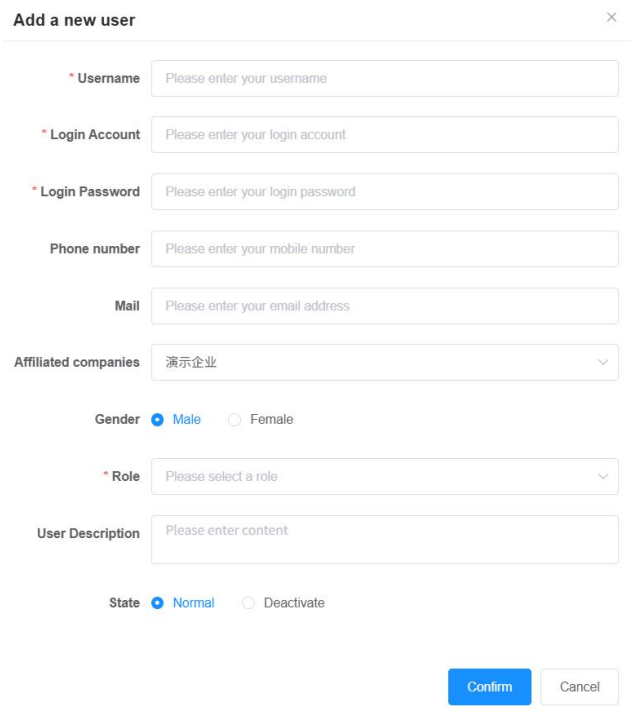
List Information in Display Area: User information is displayed in a table format:

- Serial Number: The sequential number of the user record.
- Login Account: e.g., "test2", "test1", etc.
- User Name: Records the user's name, e.g., "123", "General Employee", etc.
- Affiliated Enterprise: Displays the enterprise to which the user belongs, e.g.,

"Test", "iBEN Robot", etc.

- Phone Number: Some users have a phone number displayed, e.g., "13261990123".
- User Description: e.g., "123", "Satan", etc.
- Status: All displayed as "Normal".
- Creation Time: Records the time when the user was created, e.g., "2025-02-02 17:41:38".
- Operations: Includes "Edit", "Delete", and "Reset Password" buttons for modifying, deleting user information, or resetting the password.

Adding a New User



Mandatory Information:

- User Name: Users must enter the name of the new user.
- Login Account: Used to set the account for the user to log in to the system.

- Login Password: Sets the login password corresponding to the account.
- Role: Users must select the role corresponding to the user in the system.

Optional Information:

- Phone Number: Users can enter the user's contact phone number.
- Email: Users can enter the user's email address.
- Gender: Can be set to "Male" or "Female".
- User Description: Users can enter a brief description of the user to facilitate identification and management.
- Affiliated Enterprise: Displays "iBEN Robot" by default; users can select another enterprise via the drop-down menu (if other options are available).
- Status: Can be set to "Normal" or "Deactivated". "Normal" means the user can log in and use the system normally; "Deactivated" restricts the user's login.

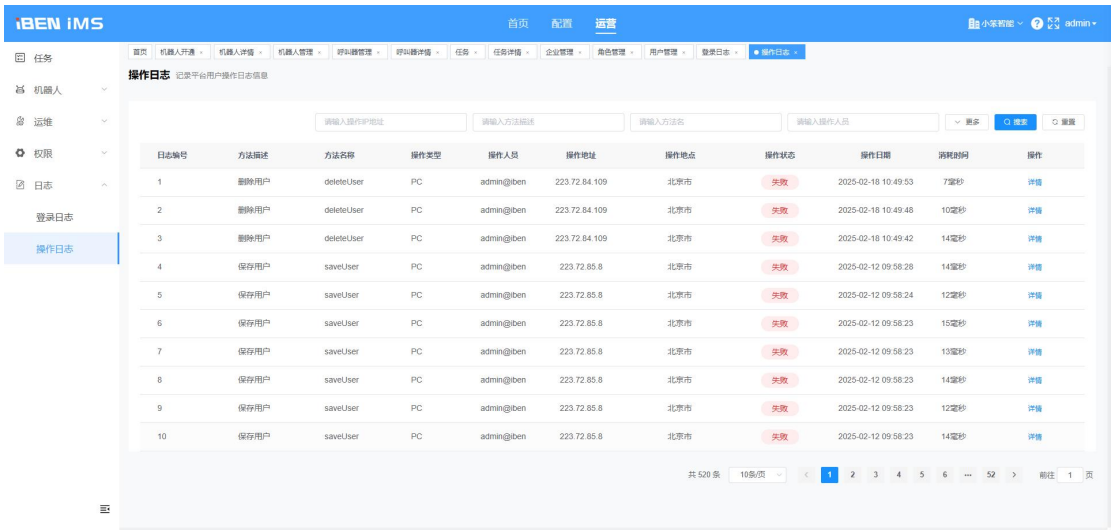
Operation Buttons:

- Confirm: Clicking this button submits the information of the new user.
- Cancel: Clicking this button cancels the operation and does not save the entered content.

2.4.5 Log

2.4.5.1 Operation Log

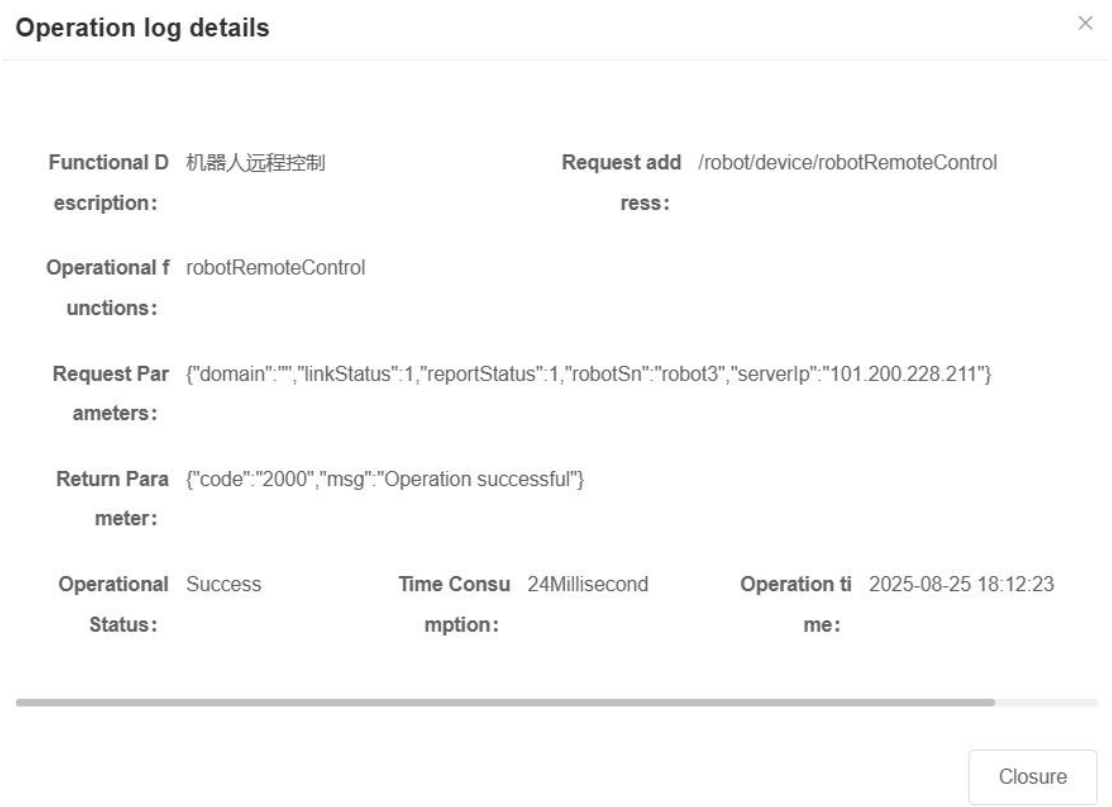
Records log information about platform user operations



Operation log-related information is presented in a table format:

- Log Number: The sequential number of each operation record.
- Method Description: e.g., "Delete User", "Save User".
- Method Name: e.g., "deleteUser", "saveUser".
- Operation Type: All displayed as "PC".
- Operator: All displayed as "admin@iben" .
- Operation Address: e.g., "223.72.84.109", "223.72.85.8".
- Operation Location: All displayed as "Beijing".
- Operation Status: All displayed records are "Failed".
- Operation Date: Records the specific operation time, e.g., "2025-02-18 10:49:53", "2025-02-12 09:58:23".
- Time Consumed: e.g., "7 ms", "10 ms", etc.
- Operations: Click "Details" to view detailed information about the operation log.

Log Details

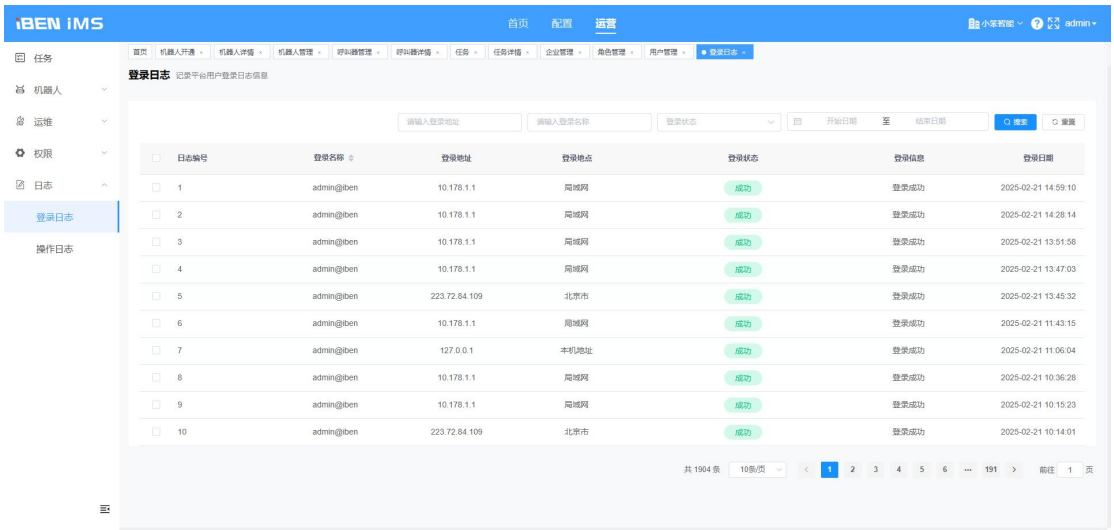


- Function Description: Clearly indicates that the operation is "Delete User".
- Request Address: Displays "robot/web/deleteUser", representing the network address of the operation request.
- Operation Function: The corresponding function method is "deleteUser".
- Request Parameters: The parameter is "[1]".
- Return Parameters: The returned information is "{"code":"10014","msg":" Super administrators cannot be deleted. Please try again after changing."}", indicating that the operation is not allowed because the user to be deleted is a super administrator.
- Operation Status: Displays "Failed".
- Time Consumed: The operation takes "7 ms".
- Operation Time: The operation occurs at "2025-02-18 10:49:53".

- Exception Information: Consistent with the return parameters, it is "{"code":"10014","msg":" Super administrators cannot be deleted. Please try again after changing."}", explaining the reason for the operation failure.
- Close Button: Clicking the "Close" button closes the pop-up window.

2.4.5.2 Login Log

Records log information about platform user logins.



Login log-related information is displayed in a table format:

- Log Number: The sequential number of each login record.
- Login Name: All displayed as "admin@iben" in this case.
- Login Address: e.g., "10.178.1.1", "223.72.84.109", etc.
- Login Location: Displays "Local Area Network", "Local Host", "Beijing", etc.
- Login Status: All displayed as "Successful".
- Login Information: Displays "Login Successful".
- Login Date: Records the specific login time, e.g., "2025-02-21 14:59:10", "2025-02-21 10:14:01", etc.